

The Emerging Terbinafine-Resistant *Trichophyton* Epidemic: What Is the Role of Antifungal Susceptibility Testing?

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Keywords

Dermatophytes · *Trichophyton* · Terbinafine · Resistance · Antifungal susceptibility testing

Abstract

Background: Dermatophytosis is commonly encountered in the dermatological clinics. The main aetiological agents in dermatophytosis of skin and nails in humans are *Trichophyton* (*T.*) *rubrum*, *T. mentagrophytes* and *T. interdigitale* (former *T. mentagrophytes*-complex). Terbinafine therapy is usually effective in eradicating infections due to these species by inhibiting their squalene epoxidase (SQLE) enzyme, but increasing numbers of clinically resistant cases and mutations in the SQLE gene have been documented recently. Resistance to antimycotics is phenotypically determined by antifungal susceptibility testing (AFST). However, AFST is not routinely performed for dermatophytes and no breakpoints classifying isolates as susceptible or resistant are available, making it difficult to interpret the clinical impact of a minimal inhibitory concentration (MIC). **Summary:** PubMed was systematically searched for terbinafine susceptibility testing

of dermatophytes on October 20, 2020, by two individual researchers. The inclusion criteria were *in vitro* terbinafine susceptibility testing of *Trichophyton* (*T.*) *rubrum*, *T. mentagrophytes* and *T. interdigitale* with the broth microdilution technique. The exclusion criteria were non-English written papers. Outcomes were reported as MIC range, geometric mean, modal MIC and MIC₅₀ and MIC₉₀ in which 50 or 90% of isolates were inhibited, respectively. The reported MICs ranged from <0.001 to >64 mg/L. The huge variation in MIC is partly explained by the heterogeneity of the *Trichophyton* isolates, where some originated from routine specimens (wild types) whereas others came from non-responding patients with a known SQLE gene mutation. Another reason for the great variation in MIC is the use of different AFST methods where MIC values are not directly comparable. High MICs were reported particularly in isolates with SQLE gene mutation. The following SQLE alterations were reported: F397L, L393F, L393S, H440Y, F393I, F393V, F415I, F415S, F415V, S443P, A448T, L335F/A448T, S395P/A448T, L393S/A448T, Q408L/A448T, F397L/A448T, I121M/V237I and H440Y/F484Y in terbinafine-resistant isolates.

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Introduction

Dermatophytes are slow-growing moulds with the ability to invade keratinised tissue such as skin, hair and nails [1]. They are classified into three groups depending on their natural habitat: geophilic (soil), zoophilic (animals) and anthropophilic (humans).

De Hoog et al. [2] revised the taxonomy of dermatophytes in 2017 and emphasised the importance of distinguishing between *Trichophyton* (*T.*) *mentagrophytes* (zoophilic strains) and *T. interdigitale* (anthropophilic strains) within the “*T. mentagrophytes* complex”. Examples of other species in this complex are for example, *T. quinckeanum* (typical source of infection: cats, dogs and mice) [3], *T. equinum* (horse) [2] and *T. erinacei* (hedgehogs) [4]. However, the revised taxonomy is not yet fully implemented. Hence, the term *T. mentagrophytes* complex is commonly used, and the specific genotype of *T. mentagrophytes* which shows alarming resistance to terbinafine (TRB) is often referred to as *T. interdigitale* [5–8]. In this context, it is important to appreciate that the names used in this paper are the original published names used in the articles included in the review which may refer to species names used before the new classification was proposed in 2017. Therefore, some published *T. interdigitale* may not be *T. interdigitale* sensu stricto but rather a specific genotype of *T. mentagrophytes*, that is, *T. mentagrophytes* ITS genotype VIII, which has recently been identified to cause a morphologically atypical, often extensive, recalcitrant form of tinea corporis [9, 10].

The dermatophytes *T. rubrum*, *T. interdigitale* and *T. mentagrophytes*, are the main aetiological agents of dermatophytosis of skin and nails in humans [11–15]. The prevalence of skin mycosis is 14–25% worldwide [16, 17], which makes it one of the most common skin diseases [12, 18]. Medical treatment of dermatophytosis consists of oral (e.g., TRB, itraconazole and griseofulvin) and/or topical antifungal agents (e.g., TRB, ketoconazole, ciclopirox, clotrimazole). TRB has been preferred as first-line agent against *Trichophyton* species because of its clinical efficacy and because TRB, in contrast to itraconazole, does not select for azole resistance in the colonising *Candida* microflora [19]. TRB is a fungicidal allylamine compound that interferes with ergosterol synthesis and thus the fungal cell membrane synthesis via its inhibition of the squalene epoxidase (SQLE) enzyme [20, 21].

TRB therapy of dermatophytosis is usually effective, yet increasing numbers of non-responding cases have been documented [22–26]. Treatment failures and recurrences can be due to host factors such as poor compliance,

drug interactions and in cases of onychomycosis local nail factors [27]. TRB resistance in dermatophytes is typically rare [28], but reports have been increasing since 2017. It is associated with various point mutations in the SQLE target gene of *T. rubrum* and *T. interdigitale* [22]. In India, TRB resistant dermatophytosis has become epidemic with common resistant or relapsing atypical and widespread ringworm infections [10, 29, 30]. Widespread use of over-the-counter topical creams containing a triple combination of antibacterial and antifungal agents, as well as a steroid may have contributed to the epidemic, as repeated exposures to sub-inhibitory concentrations of antifungals in an immunosuppressed environment caused by the steroid increases the risk of resistance [31–33]. However, cases of TRB-resistant *Trichophyton* infections are also seen in other Asian and Middle Eastern countries such as Japan and Iran [21, 34, 35]. In Europe, resistant strains of *T. rubrum* and *T. interdigitale* with SQLE mutations have been identified in Switzerland [25], Germany [36, 37], Poland [38], Belgium [39] and Denmark [22–24].

Clinical Signs of Antifungal Drug Resistance

It would be ideal if the term “Treatment resistant dermatophytosis” was used exclusively to describe clinically resistant cases in which antifungal resistance had been confirmed both *in vivo* and *in vitro* either through susceptibility testing (demonstrating a high MIC) or molecularly (demonstrating a recognised resistance mutation). It is not possible in all countries to follow such a stringent definition. What is seen instead is “difficult to treat infections” with persistent superficial dermatophytosis including chronic (more than 6 months) and recurrent (re-appearance of lesions 4 weeks after cessation of standard recommended treatment after clinical cure) forms with poor or no response to standard recommended treatment [31].

Certain clinical characteristics are useful indicators of drug resistance. The most important is a widespread dermatophytosis that has been previously treated, often with multiple regimens. These patients often present with large, sometimes coalescing lesions of superficial dermatophytosis with severe itching. In these cases, the lesions can range from highly inflammatory ones that are raised with erythema and variable scaling, to minimally inflammatory ones with uniform small whitish appearing scales. Most of the patients have concomitant tinea corporis and tinea cruris which often extends posteriorly to the perineum, posterior thighs and buttocks (Fig. 1). Tinea faciei often develops secondarily in these patients, at times ex-



Fig. 1. Widespread dermatophytosis caused by *T. mentagrophytes* ITS genotype VIII.

tending from the neck to the face, including ears, and in some cases to the scalp, but usually not very far inside the hair margins. Occasionally, TRB resistance is seen in patients with a localised infection, which do not respond to treatment regimens that would normally cure the infection (Fig. 2).

Antifungal Susceptibility Testing

Resistance to antifungal compounds is determined by antifungal susceptibility testing (AFST). A micro-organism is defined as clinically susceptible (S) by a level of antimicrobial activity associated with a high likelihood of therapeutic success using a standard dosing regimen of the agent, whereas it is considered clinically resistant (R) when therapeutic failure is expected [40]. Various methods have been adopted for the testing of yeast or moulds including disk diffusion, agar dilution, E-test, flow cytometry and broth micro- and macrodilution techniques



Fig. 2. Localised terbinafine resistant *T. rubrum* confirmed by EUCAST method E.Def 11.0 (45).

with or without a colour indicator [31, 41]. AFST of dermatophytes is not routinely performed because it is complicated by the slow growth of the fungi and the associated risk of contamination, because no commercial susceptibility tests are validated for dermatophyte testing, because interpretative clinical breakpoints that allow interpretation of the specific MIC as susceptible or resistant have not been established, and because the false perception of low resistance rates creates an assumption that testing is not needed. Numerous critical parameters such as inoculum size, incubation time, type of microdilution plate, choice of medium and endpoint determination affect susceptibility test results. Currently, there are only two standardised techniques for *in vitro* AFST of dermatophytes; one from the Clinical Laboratory Standards Institute (CLSI) and the other from the European Committee on Antimicrobial Susceptibility Testing (EUCAST) [42, 43]. Both of these reference methods are performed in microtitre plates containing broth culture media with serial dilutions of antifungal drugs, wherein a standardised amount of dermatophyte cells are added and the fungus growth is compared with positive and negative controls. The main differences of the two methods are that the CLSI method recommends a visual no-growth endpoint in round bottom plates, whereas the EUCAST adopted flat-bottom plates and an automated spectrophotometer reading to avoid subjectivity in endpoint determination. Moreover, the EUCAST has recently estab-

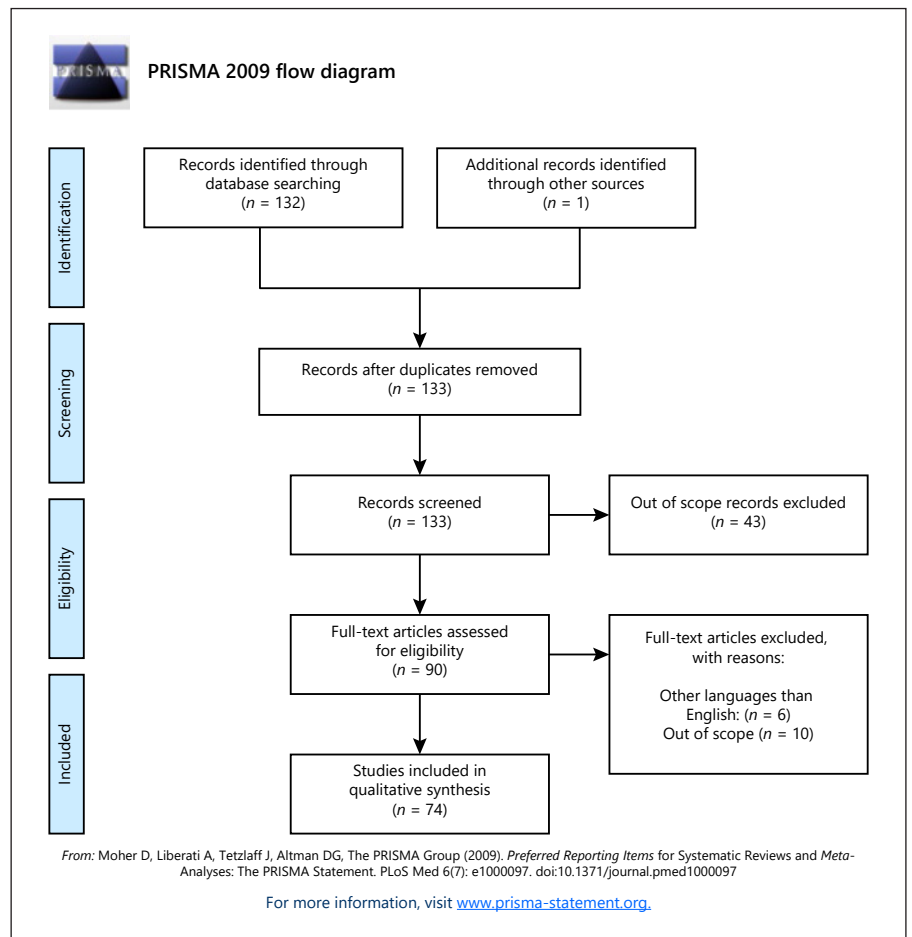


Fig. 3. PRISMA flow diagram of the literature search.

lished tentative epidemiological cut off values (ECOFFs, defined as the upper limit of the wild-type population [44]) that allows classification of the tested organism as either wild type (normal, and thus you can trust your clinical experience with respect to efficacy of medical treatment) or non-wild type (less susceptible than normal, and thus the dermatophyte may not respond as well to the typical therapy).

Despite the lack of routine susceptibility testing, increasing numbers of clinically resistant dermatophytes due to isolates with high TRB MICs and with target gene mutations have been reported in the past few years. This review evaluates the reported MICs for *T. rubrum*, *T. mentagrophytes* and *T. interdigitale* with the broth microdilution method in the literature and addresses the problem of extrapolation from the laboratory bench to patients.

Method

This systematic review was conducted in accordance with the PRISMA guidelines [45] (Fig. 3)

PubMed was searched for TRB susceptibility testing of *T. rubrum*, *T. mentagrophytes* and *T. interdigitale* with the following search string: ("disease susceptibility"[MeSH Terms] OR ("disease"[All Fields] AND "susceptibility"[All Fields]) OR "disease susceptibility"[All Fields] OR "susceptibility"[All Fields]) AND ("terbinafine"[MeSH Terms] OR "terbinafine"[All Fields]) AND ("trichophyton"[MeSH Terms] OR "trichophyton"[All Fields]). The literature search and screening were conducted on October 20, 2020, by two individual researchers (J.J.S. and D.M.L.S.). The inclusion criteria were *in vitro* terbinafine susceptibility testing of human isolates of *T. rubrum*, *T. mentagrophytes* and *T. interdigitale* with the broth microdilution technique. The exclusion criteria were papers in languages other than English. See Figure 1 for additional details of the literature search. Data from the included papers were extracted in a predefined Excel chart, and outcomes were reported as MIC range, MIC in which 50% of isolates were inhibited (MIC₅₀), MIC in which 90% of isolates were inhibited (MIC₉₀), geometric mean (GM-MIC), most frequent MIC (Modal MIC), method, number of isolates and test material. Whenever data was not available this was stated as NA.

Results

T. interdigitale

TRB susceptibility of a total of 829 *T. interdigitale* isolates were reported between 2001 and 2020 in 24 studies by the following reference standards: CLSI-M38-A2 (19 studies), EUCAST E.Def 11.0 (2 studies), EUCAST E.Def 9.3.1 (1 study), EUCAST E.Def 9.2 (1 study), CLSI-M38-A (1 study) and CLSI-M38-P/NCCLS-M38-P (1 study) (Table 1). For all *T. interdigitale* isolates, the MIC ranged from 0.001 to >32 mg/L (21 studies) using the CLSI/NCCLS method and from <0.008 to >8 mg/L (4 studies) using the EUCAST method. Details of MIC₅₀, MIC₉₀, GM-MIC and modal MIC are available in Table 1. Overall, 85 isolates were from patients with clinical failures, and two isolates were deemed resistant because of high MIC and SQLE mutations. For *T. interdigitale* isolates from clinically non-responding patients, the MIC ranged from <0.015 to >32 mg/L and <0.008 to >8 using the CLSI/NCCLS method (6 studies) and EUCAST method (2 studies), respectively. Six studies performed SQLE gene sequencing detecting the following SQLE mutations: F397L, L393F and F397L/A448T. Three studies proposed the following breakpoints (susceptible [S] and resistant [R]) for TRB against *T. interdigitale* strains using CLSI reference standard: R ≥1 mg/L [46], R ≥4 mg/L [12, 47], and one study also using CLSI as reference proposed high TRB MIC at ≥2 mg/L [11]. One study [48] proposed the following wild-type upper limits (WT-UL) using EUCAST as reference standard: WT-UL ≤0.25 mg/L (by visual endpoint reading and spectrophotometric endpoint reading with a 90% growth inhibition [spec-90%]), WT-UL ≤0.125 mg/L (by spectrophotometric endpoint reading with a 50% growth inhibition [spec-50%]). The spec-50% WT-UL values have subsequently been accepted as EUCAST tentative ECOFF for the EUCAST 11.0 method [43].

T. mentagrophytes

TRB susceptibility of a total of 1,399 *T. mentagrophytes* isolates were reported between 1995 and 2020 in 41 studies by the following reference standards: CLSI-M38-A2 (18 studies), CLSI-M38-A/NCCLS-M38-A (10 studies), CLSI-M38-P/NCCLS-M38-P (5 study), CLSI-M38 (1 study), CLSI-M27-A/NCCLS-M27-A (3 studies), EUCAST E.Def 9.3.1 (1 study), EUCAST E.Dis 7.1 (1 study) and unspecified broth microdilution method (2 studies) (Table 1). For all *T. mentagrophytes* isolates, the MIC ranged from 0.001 to >32 mg/L using the CLSI/NCCLS method (37 studies) and <0.007 to 4 mg/L using EUCAST method (2 studies) (Table 1). For *T. mentagrophytes* iso-

lates with clinical resistance, the MIC ranged from 0.015 to >32 mg/L using the CLSI/NCCLS method (5 studies), and one EUCAST study reported TRB MIC 4 mg/L [39]. Overall, 36 isolates were from patients with clinical failures, and 210 isolates were suspected as clinically resistant because of high MIC and SQLE mutations. Five studies performed SQLE gene sequencing detecting the following SQLE mutations: F397L, L393F, L393S, H440Y, S443P, A448T, L335F/A448T, S395P/A448T, L393S/A448T, Q408L/A448T and F397L/A448T. Two studies using the CLSI reference method defined TRB resistance as R ≥1 mg/L [9] and R ≥4 mg/L [47].

T. mentagrophytes-T. interdigitale complex

TRB susceptibility of a total of 548 isolates reported as belonging to the *T. mentagrophytes-T. interdigitale* complex were reported between 2019 and 2020 in 4 studies by the CLSI-M38-A2 reference standard (Table 1). For all *T. mentagrophytes-T. interdigitale* complex isolates, the MIC ranged from 0.015 to 32 mg/L (4 studies). Overall, 16 isolates were from patients with clinical failures, and 74 isolates were suspected as clinically resistant because of high MIC and SQLE mutations. The MIC ranged from 1 to 32 mg/L (3 studies). Three studies that performed SQLE gene sequencing detected the following SQLE mutations: F397L and Q408L. One study proposed WT-UL ≤8 mg/L [5], and one study proposed high TRB MIC at >2 mg/L [6] both using CLSI as reference standard.

T. rubrum

In the years between 1995 and 2020, 62 out of the 74 included studies tested TRB susceptibility of *T. rubrum* with the following standards: CLSI-M38-A2 (31 studies), CLSI-M38-A/NCCLS-M38-A (13 studies), CLSI-M38-P/NCCLS-M38-P (8 studies), CLSI-M27-A/NCCLS-M27-A (4 studies) EUCAST E.Def 11.0 (2 studies), EUCAST E.Def 9.3.1 (1 study), EUCAST E.Def 9.2 (1 study), EUCAST E.Dis 7.1 (1 study) and unspecified broth microdilution method (3 studies) (Table 2). A total of 2,235 *T. rubrum* isolates were included, of which 113 isolates were from patients with clinical resistance, and 26 isolates were suspected as clinically resistant because of high MICs and SQLE mutations. The MICs for all *T. rubrum* isolates ranged from <0.001 to >64 mg/L using the CLSI/NCCLS method (56 studies) and <0.008 to >8 mg/L using the EUCAST method (5 studies) (Table 2). For *T. rubrum* isolates with clinical resistance, the MIC ranged from 0.003 to >32 mg/L using the CLSI/NCCLS method (8 studies) and 0.03 to >8 mg/L using the EUCAST method (2 studies). Two studies performed SQLE gene sequencing de-

Table 1. Summary of published susceptibility testing data for isolates of *Trichophyton (T.) interdigitale*, *T. mentagrophytes* and *T. mentagrophytes-T. interdigitale* complex

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC50, mg/L	GM-MIC, mg/L	Modal MIC, mg/L	Detected SQLE mutations	Material
<i>T. interdigitale</i> isolates								
<i>CLSI/NCCLS method: all isolates</i>								
Taghipour [21], Iran, 2020	CLSI M38-A2	96	0.03–0.25	0.062	0.125	0.038	NA	Clinical isolates
Salehi [14], Iran, 2020	CLSI M38-A2	48	0.003–0.25	0.03	0.06	0.015	NA	Clinical isolates
Behnam [66], Iran, 2020	CLSI M38-A2	18	0.002–1	0.125	0.5	NA	NA	Skin, hair, nails
Allahdadi [67], Iran, 2019	CLSI M38-A2	2	0.004–0.032	0.008	0.032	NA	NA	Hair
Hiruma [34], Japan, 2019	CLSI M38-A2	Susceptible 23 Resistant 1	<0.03	<0.03	<0.03	NA	NA	Nails, skin
Pathania [47], India, 2018	CLSI M38-A2		0.015–8	0.03	8	NA	NA	Skin
Khurana [46], India, 2018	CLSI M38-A2	64	0.25–≥32	1	32	2.813	NA	Skin
Salehi [20], Iran, 2018	CLSI M38-A2	27	0.003–0.125	0.01	0.06	0.01	NA	Skin
Singh [12], India, 2018	CLSI M38-A2	63	0.06–32	1	32	2.7	F397L L393F	Skin, nails, hair
Rudramurthy [11], India, 2018	CLSI M38-A2	88	0.015–32	0.03	4	0.06	F397L	Skin
Rezaei-Matehkolaie [68], Iran, 2018	CLSI M38-A2	66	0.004–0.12	0.016	0.03	0.013	NA	Skin, nails, hair
Carreira [69], Portugal, 2018	CLSI M38-A2	14	mPDB medium 0.0015–0.004 RPMI 1640 0.001–0.031	NA	NA	NA	NA	Skin
Altınbaş [15], Turkey, 2018	CLSI M38-A2	2	0.06–0.06	NA	NA	0.06	NA	Skin, nails, hair
Dabas [70], India, 2017	CLSI M38-A2	37	0.03–16	0.5	1	0.375	NA	Skin
Yamada [25], Switzerland, 2017	CLSI M38-A2	1	3.2	NA	NA	NA	F397L	Clinical isolates
Afshari [71], Iran, 2016	CLSI M38-A2	12	0.0313–16	0.5	4	1.49	NA	Skin
Ansari [72], Iran, 2015	CLSI M38-A2	156	0.004–0.25	0.016	0.125	0.017	NA	Skin, nails, hair
Tamura [73], Japan, 2014	CLSI M38-A2	14	0.03–0.06	0.03*	0.03*	NA	NA	Skin, nails
Silva [74], Brazil, 2014	CLSI M38-A2	24	≤0.031	≤0.031	≤0.031	0.03	NA	Skin, nails, hair
Magagnin [75], Brazil, 2011	CLSI M38-A	4	0.03–0.06	NA	NA	0.04	NA	Clinical isolates
Fernández-Torres [65], Spain, 2001	NCCLS-M38-P	21	0.007–2	0.01	0.06	0.01	NA	Clinical isolates
<i>CLSI/NCCLS method: isolates from clinically nonresponding patients</i>								
Hiruma [34], Japan, 2019	CLSI M38-A2	1 ^a	2	NA	NA	NA	NA	Nails, skin
Pathania [47], India, 2018	CLSI M38-A2	10	0.015–8	0.03	8	NA	NA	Skin

Table 1 (continued)

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC ₅₀ , mg/L	MIC ₉₀ , mg/L	GM-MIC, mg/L	Modal MIC, mg/L	Detected SQLE mutations	Material
Khurana [46], India, 2018	CLSI-M38-A2	28	0.5–≥32	8	≥32	NA	NA	F397L L393F	Skin
Singh [12], India, 2018	CLSI M38-A2	20	4–≥32	≥32	≥32	NA	≥32	F397L L393F	Skin, nails, hair
Rudramurthy [76], India, 2018	CLSI M38-A2	15	2–32	16*	32*	9.62*	16*	F397L	Skin
Yamada [25], Switzerland, 2017	CLSI M38-A2	1 ^a	3.2	NA	NA	NA	NA	F397L	Clinical isolates
<i>EUCAST method: all isolates</i>									
Markantonatou [77], Greece, 2020	EUCAST E.Def 9.3.1	8	<0.031	<0.031	<0.031*	NA	<0.031*	NA	Clinical isolates
Arendrup [48], Denmark, 2020	EUCAST E.Def 11.0	WT 10	Visual reading 0.008–0.5 Spec-90% <0.008–0.25 Spec-50% <0.008–125 0.016	0.06	NA	NA	0.06	NA	Clinical isolates
		Mutant 10	Visual reading 0.016–>8 Spec-90% <0.008–>8 Spec-50% <0.008–>8	4 4 4 2	NA	NA	8 4 2	F397L L393F F397L/A448T	
Saunte [22], Denmark, 2019	EUCAST E.Def 11.0	WT 8	Without CC 0.008–0.016 With CC	0.016*	0.016*	0.012*	0.016*	NA	Clinical isolates
		Mutant 2	Without CC >8 With CC >4–>8	0.016*	0.03*	0.017*	0.016*	F397L L393F	
Dabas [70], India, 2017	EUCAST E.Def 9.2	37	0.06–16	1	2	0.683	1.0	NA	Skin
<i>EUCAST method: isolates from clinically non-responding patients</i>									
Arendrup [48], Denmark, 2020	EUCAST. E.Def 11.0	10	Visual reading 0.016–>8 Spec-90% <0.008–>8 Spec-50% <0.008–>8	4 4 4 2	NA	NA	8 4 2	F397L L393F F397L/A448T	Clinical isolates
Saunte [22], Denmark, 2019	EUCAST E.Def 11.0	2	Without CC >8 With CC >4–>8	NA	NA	NA	NA	F397L L393F	Clinical isolates
<i>T. mentagrophytes</i>									

Table 1 (continued)

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC ₅₀ , mg/L	MIC ₉₀ , mg/L	GM-MIC, mg/L	Modal MIC, mg/L	Detected SQLE mutations	Material
<i>CLSI/NCCLS method: all isolates</i>									
Taghipour [21], Iran, 2020	CLSI M38-A2	45	0.007–>32	0.0625	>32	0.12	NA	F397L/A448T L393S/A448T	Clinical isolates
Ebert [29], Germany/Indian isolates, 2020	CLSI M38-A2	279	0.125–16	NA	NA	NA	8*	F397L L393F L393S H440Y S443P A448T L335F/A448T S395P/A448T F397L/A448TQ408L/A448T	Skin
Gnat [78], Poland, 2020	CLSI M38-A2	17	0.004–2	1	2	0.91	0.004	NA	Skin, hair, nails
Behnam [66], Iran, 2020	CLSI M38-A2	21	0.002–1	0.5	0.5	NA	NA	NA	Skin, hair, nails
Łagowski [38], Poland, 2020	CLSI M38-A2	7	0.004–32	0.004	0.016	2.21	0.004	L393F	Skin, hair
Fattahi [35], Iran, 2020	CLSI M38	4	≥8	NA	NA	NA	NA	F397L	Skin
Sardana [18], India, 2018	CLSI M38-A2	14	NA	NA	0.125	NA	NA	NA	Skin
Pathania [47], India, 2018	CLSI M38-A2	36	0.015–8	0.125	8	NA	NA	NA	Skin, nails
Filho [79], Brazil, 2017	CLSI M38-A2	15	≤0.12	NA	NA	NA	NA	NA	Nail
Oz [80], Turkey, 2017	CLSI M38-A2	9	0.0019–0.0039	0.0039	0.0039	NA	NA	NA	Skin, nails, hair
Muangkaew [81], Thailand, 2017	CLSI M38-A2	1	0.06	NA	NA	NA	NA	NA	Skin
Bhatia [82], India, 2015	CLSI M38-A2	34	0.625–4	0.5	2	NA	NA	NA	Skin, nails, hair
Badali [83], Iran, 2015	CLSI M38-A2	13	0.016–0.31	0.031	0.31	0.03	NA	NA	Skin, nails, hair
Yenişehirli [84], Turkey, 2013	CLSI M38-A2	49	0.015–0.125	0.06	0.125	0.06	NA	NA	Skin, nails, hair
Adimi [85], Iran, 2013	CLSI M38-A2	136	0.0156–16	0.0312	16	0.142	NA	NA	Skin, nails, hair
Jo Siu [86], USA, 2013	CLSI M38-A2	129	0.004–0.5	0.008	0.03	0.010	NA	NA	Nails
Coulibaly [87], France, 2013	CLSI M38-A2	3	<0.25	NA	NA	NA	NA	NA	Hair
Irimie [88], Romania, 2013	CLSI M38-P	ESRD 4 Control 4	NA	0.015	0.031	NA	NA	NA	Skin, nails
Magagnin [75], Brazil, 2011	CLSI M38-A	8	0.03–16	0.03*	16*	0.04	0.03*	NA	Clinical isolates

Table 1 (continued)

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC ₅₀ , mg/L	MIC ₉₀ , mg/L	GM-MIC, mg/L	Modal MIC, mg/L	Detected SQLE mutations	Material
Ozcan [89], Turkey, 2010	CLSI M-38-P	NIDDM 11 Controls 7	NA	<0.008	0.03	0.015	NA	NA	Skin, nails
Zhang [90], China, 2010	CLSI M38-A2	14	0.157–1	0.0313	1	NA	NA	NA	Skin
Mota [91], Brazil, 2009	CLSI M38-A	14	0.03–0.5	0.06	0.25	NA	NA	NA	Skin, nails
Almeida [63], Brazil, 2009	CLSI M38-A	35	0.12 – ≥16	NA	NA	NA	NA	NA	Skin, nails, hair
Araújo [92], Brazil, 2009	CLSI M38-A	14	0.03–0.5	0.06	0.25	0.08	NA	NA	Skin, nails
Koga [93], Japan, 2009	CLSI M38-A	10	0.002–0.031	0.016	0.016	0.01	NA	NA	Clinical isolates
Shehata [62], Egypt and USA, 2008	CLSI M38-A2	15	0.008–0.06	0.015	0.06	NA	NA	NA	Clinical isolates
Singh [94], Canada, 2007	CLSI M38-A	16	0.001–0.002	NA	NA	0.0014	NA	NA	Nails
Silva Barros [95], Brazil, 2007	CLSI M38-A	50	NA	0.007	0.015	NA	NA	NA	Nails
da Silva Barros [96], Brazil, 2006	CLSI M38-A	50	<0.007–0.015	0.007*	0.015*	NA	0.015*	NA	Nails
Esteban [97], Spain, 2005	NCCLS-M38-A	9	Incubation 3 days 0.0078–0.06 7 days 0.01–0.125	0.01	0.03	0.015	NA	NA	Clinical isolates
Cetinkaya [64], Turkey, 2005	NCCLS-M38-A	11	0.03–1	0.125	1	NA	NA	NA	Skin, nails
Alió [98], Venezuela, 2005	NCCLS-M38-P	10	0.1–5	NA	NA	NA	NA	NA	Skin
Gupta [99], Canada, 2003	NCCLS-M27-A	14	0.003–0.5	0.003	0.015	0.005	NA	NA	Clinical isolates
Fernández-Torres [65], Spain, 2001	NCCLS-M38-P	122	0.007–0.5	0.06	0.06	0.04	NA	NA	Clinical isolates
Jessup [100], USA, 2000	NCCLS M27-A	32	NA	0.001	0.001	NA	NA	NA	Clinical isolates
Jessup [101], USA, 2000	NCCLS-M38-P	30	0.001–0.002	NA	NA	NA	NA	NA	NA
Norris [102], Ohio, USA, 1999	NCCLS-M27-A	6	<0.06	NA	NA	NA	NA	NA	Clinical isolates
<i>CLSI/NCCLS method: isolates from clinically non-responding patients</i>									
Taghipour [21], Iran, 2020	CLSI M38-A2	5a	>32	NA	NA	NA	NA	F397L/A448T L393S/A448T	Clinical isolates
Ebert [29], Germany/India, 2020	CLSI M38-A2	199a	0.25–16	NA	NA	NA	8*	F397L L393F L393S H440Y S443P S395P/ A448T F397L/A448TQ408L/ A448T	Skin

Table 1 (continued)

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC50, mg/L	MIC90, mg/L	GM-MIC, mg/L	Modal MIC, mg/L	Detected SQLE mutations	Material
Łagowski [38], Poland, 2020	CLSI M38-A2	1a	32	NA	NA	NA	NA	L393F	Skin, hair
Fattahi [35], Iran, 2020	CLSI M38	4a	≥8	NA	NA	NA	NA	F397L	Skin
Pathania [47], India, 2018	CLSI M38-A2	36	0.015–8	0.125	8	NA	NA	NA	Skin, nails
<i>EUCAST method: all isolates</i>									
Sacheli [39], Belgium, 2020	EUCAST E.Def 9.3.1	5	0.016–4	NA	NA	NA	0.03*	F397L	Skin, hair
da Silva Barros [96], Brazil, 2006	EUCAST E.Dis 7.1	50	<0.007–0.031	<0.007*	0.015*	NA	<0.007*	NA	Nails
<i>EUCAST method: isolates from clinically non-responding patients</i>									
Sacheli [39], Belgium, 2020	EUCAST E.Def 9.3.1	1a	4	NA	NA	NA	NA	F397L	Skin, hair
<i>Microdilution method: all isolates</i>									
Santos [103], Brazil, 2006	Microdilution	30	<0.031	<0.031	<0.031	NA	NA	NA	Nails
Korting [104], Germany, 1995	Microdilution	16	0.001–0.01	NA	NA	NA	0.001*	NA	Nails
<i>T. mentagrophytes-T. interdigitale complex</i>									
<i>CLSI/NCCLS method: all isolates</i>									
Shaw [5], India, 2020	CLSI M38-A2	498	0.015–32	0.03	8	0.096	0.015	F397L	Skin, hair, nails
Shankarnarayan [6], India, 2020	CLSI M38-A2	15	2–16	8*	16*	5.78*	8*	F397L	Skin
Hsieh [7], Switzerland, 2019	CLSI M38-A2	1	≥1	NA	NA	NA	NA	Q408L	Skin
Poojary [8] India, 2019	CLSI M38-A2	34	0.016–0.256	NA	NA	NA	NA	NA	Skin, nails, hair
<i>CLSI/NCCLS method: isolates from clinically non-responding patients</i>									
Shaw [5], India, 2020	CLSI M38-A2	74a	1–32	NA	NA	NA	NA	F397L	Skin, hair, nails
Shankarnarayan [6], India, 2020	CLSI M38-A2	15	2–16	8*	16*	5.78*	8*	F397L	Skin
Hsieh [7], Switzerland, 2019	CLSI M38-A2	1	≥1	NA	NA	NA	NA	Q408L	Skin

MIC, minimal inhibitory concentration; MIC50, MIC in which 50% of isolates were inhibited; MIC90, MIC in which 90% of isolates were inhibited; G_{mean} , geometric mean; CLSI, Clinical Laboratory Standards Institute; NCCLS, National Committee for Clinical Laboratory Standards (former name of CLSI); NA, not available; WT, wild type; spec-90%/spec-50%, plate reading by spectrophotometer using 90%/50% inhibition endpoint; CC, medium supplemented with chloramphenicol & cycloheximide; NIDDM, non-insulin-dependent diabetes mellitus; mPDB medium, modified potato dextrose broth medium; RMPI 1640, Roswell Park Memorial Institute 1640. * Calculated in this study. ^a Suspected resistance because of high MIC, and/or SQLE, mutation.

Table 2. Published susceptibility testing data of *Trichophyton rubrum* isolates

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC ₅₀ , mg/L	MIC ₉₀ , mg/L	GM-MIC, mg/L	Modal MIC, mg/L	MIC for isolate with clinical resistance	Material
<i>CLSI/NCCLS method: all isolates</i>									
Salehi [14], Iran, 2020	CLSI M38-A2	10	0.003–0.25	0.125	0.25	0.0678	0.125	NA	Clinical isolates
Ebert [29], Germany/Indian isolates, 2020	CLSI M38-A2	18	0.0312–8	0.0312*	8*	NA	0.0312*	F397L	Skin
Poojary [8], India, 2019	CLSI M38-A2	36	0.008–0.256	NA	NA	NA	NA	NA	Skin, nails, hair
Allahdadi [67], Iran, 2019	CLSI M38-A2	1	NA	0.016	0.032	NA	NA	NA	Hair
Carreira [69], Portugal, 2018	CLSI M38-A2	15	mPDB medium 0.002–0.004 RPMI 1640 0.0015–0.031	NA	NA	NA	NA	NA	Skin
Salehi [20], Iran, 2018	CLSI M38-A2	20	0.003 – >32	0.03	1.3	0.04	NA	L393F	Skin
Sardana [18], India, 2018	CLSI M38-A2	11	NA	NA	0.125	NA	NA	NA	Skin
Pathania [47], India, 2018	CLSI M38-A2	29	0.015–16	0.063	8	NA	NA	NA	Skin, nails
Rudramurthy [76], India, 2018	CLSI M38-A2	35	0.015–16	0.015	2	0.05	NA	F397L	Skin
Rezaei-Matehkolaei [68], Iran, 2018	CLSI M38-A2	54	0.004–0.06	0.008	0.016	0.009	0.008	NA	Skin, nails, hair
Altunbaş [15], Turkey, 2018	CLSI M38-A2	55	0.06–0.06	0.06	0.06	0.06	NA	NA	Skin, nails, hair
Oz [80], Turkey, 2017	CLSI M38-A2	35	0.0019–1	0.078	0.078	NA	NA	NA	Skin, nails, hair
Filho [79], Brazil, 2017	CLSI M38-A2	28	≤0.06	NA	NA	NA	NA	NA	Nail
Yamada [25], Switzerland, 2017	CLSI M38-A2	12	0.1 – >12.8	3.2*	6.4*	NA	6.4*	F397L L393F L393S F393I F393V F415I F415S F415V H440Y	Clinical isolates
Dabas [70], India, 2017	CLSI M38-A2	5	0.03–8	0.5	8	0.863	0.5	NA	Skin
Muangkaew [81], Thailand, 2017	CLSI M38-A2	2	≤0.06–0.06	NA	NA	NA	NA	NA	Skin
Alshari [71], Iran, 2016	CLSI M38-A2	5	0.0313–16	2	4	1.49	1.097	NA	Skin
Nizam [105], Malaysia, 2016	CLSI M38-A2	3	2–4	NA	NA	3.17	NA	NA	Dermatological specimen
Bhatia [82], India, 2015	CLSI M38-A2	18	0.03–4	0.5	2	NA	NA	NA	Skin, nails, hair
Badali [83], Iran, 2015	CLSI M38-A2	13	0.016–0.31	0.031	0.31	0.04	NA	NA	Skin, nails, hair
Ansari [72], Iran, 2015	CLSI M38-A2	60	0.004–0.125	0.016	0.063	0.017	NA	NA	Skin, nails, hair
Azambuja [106], Brazil, 2014	CLSI M38-A2	12	0.0078–>0.5	0.0625	0.125	NA	NA	NA	Nails
Tamura [73], Japan, 2014	CLSI M38-A2	14	0.004–0.06	0.03*	0.03*	NA	NA	NA	Skin, nails, hair
Silva [74], Brazil, 2014	CLSI M38-A2	37	≤0.031–0.062	≤0.031	≤0.031	0.06	NA	NA	Skin, nails, hair

Table 2 (continued)

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC50, mg/L	MIC90, mg/L	GM-MIC, mg/L	Modal MIC, mg/L	MIC for isolate with clinical resistance	Material
Yenişehirli [84], Turkey, 2013	CLSI M38-A2	78	0.007–0.5	0.03	0.125	0.04	NA	NA	Skin, nails, hair
Irimie [88], Romania, 2013	CLSI M38-P	ESRD 22 Control 23	NA	0.007	0.015	NA	NA	NA	Skin, nails
Adimi [85], Iran, 2013	CLSI M38-A2	89	0.0156–16	0.0312	16	0.172	NA	NA	Skin, nails, hair
Jo Siu [86], USA, 2013	CLSI M38-A2	130	0.004–0.06	0.008	0.015	0.009	NA	NA	Nails
Coulibaly [87], France, 2013	CLSI M38-A2	3	<0.25	NA	NA	NA	NA	NA	Hair
Ghannoun [49], USA, 2011	CLSI M38-A2	Susceptible 20 Resistant 5	0.015–0.03 2–8	0.015 NA	0.03 NA	NA	NA	NA	Clinical isolates
Magagnin [75], Brazil, 2011	CLSI M38-A	3	0.03–0.06	NA	NA	0.07	NA	NA	Clinical isolates
Ozcan [89], Turkey, 2010	CLSI M-38-P	NIDDM 44 Controls 43	NA	<0.008 <0.008	0.015 <0.008	0.018 0.008	NA	NA	Skin, nails
Zhang [90], China, 2010	CLSI M38-A2	14	0.031–0.125	0.0625	0.125	NA	NA	NA	Skin
Mota [91], Brazil, 2009	CLSI M38-A	27	0.03–0.5	0.125	0.25	NA	NA	NA	Skin, nails
Koga [93], Japan, 2009	CLSI M38-A	10	0.002–0.0078	0.0039	0.0078	0.0037	NA	NA	Clinical isolates
Almeida [63], Brazil, 2009	CLSI M38-A	21	0.12–≥16	NA	NA	NA	NA	NA	Skin, nails, hair
Araújo [92], Brazil, 2009	CLSI M38-A	27	0.03–0.5	0.125	0.25	0.11	NA	NA	Skin, nails
Shehata [62], Egypt and USA, 2008	CLSI M38-A2	15	0.004–>64	0.03	2	NA	NA	NA	Clinical isolates
Coelho [107], Brazil, 2008	CLSI M38-A	2	0.20	NA	NA	NA	NA	NA	Clinical isolates
Singh [94], Canada, 2007	CLSI M38-A	16	0.001–0.03	NA	NA	0.006	NA	NA	Nails
Silva Barros [95], Brazil, 2007	CLSI M38-A	50	NA	<0.007	0.007	NA	NA	NA	Nails
Sarıfakioglu [108], Turkey, 2007	CLSI M38-P	39	≤0.008–0.015	<0.008	0.008	NA	NA	NA	Nails
de Assis Santos [109], Brazil, 2007	CLSI M38-A	40	<0.031	<0.031*	<0.031*	NA	<0.031*	NA	Nails
da Silva Barros [96], Brazil, 2006	CLSI M38-A	50	<0.007–0.031	<0.007*	0.007*	NA	<0.007*	NA	Nails
Santos [11], Brazil, 2005	CLSI M38-A	50	<0.031	<0.031*	<0.031*	NA	NA	NA	Clinical isolates
Esteban [97], Spain, 2005	NCCLS-M38-A	10	Incubation 3 days 0.0039–0.06 7 days 0.0078–0.06	0.0078	0.01	0.008	NA	NA	Clinical isolates

Table 2 (continued)

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC ₅₀ , mg/L	MIC ₉₀ , mg/L	GM-MIC, mg/L	Modal MIC, mg/L	MIC for isolate with clinical resistance	Material
Cetinkaya [64], Turkey, 2005	NCCLS-M38-A	108	0.03–4	0.06	0.25	NA	NA	NA	Skin, nails
Alió [98], Venezuela, 2005	NCCLS-M38-P	10	0.1–1	NA	NA	NA	NA	NA	Skin
Gupta [110], Canada, 2003	NCCLS-M27-A	18	0.003–2	0.003	0.003	0.003	0.003*	NA	Nails
Gupta [99], Canada, 2003	NCCLS-M27-A	68	0.003–1	0.003	0.003	0.003	NA	NA	Clinical isolates
Fernández Torres [111], Spain, 2002	NCCLS-M38-P	100	0.0039–0.25	NA	NA	0.01	NA	NA	Clinical isolates
Fernández-Torres [65], Spain, 2001	NCCLS-M38-P	144	0.003 – >16	0.01	0.06	0.01	NA	NA	Clinical isolates
Fernández-Torres [112], Spain, 2000	NCCLS-M38-P	100	Inoculum transmission 80–85% 0.0039–0.25 65–70% 0.0039–0.25	0.0078 0.01	0.03 0.01	0.0098 0.01	NA	NA	Skin, nails
Jessup [100], USA, 2000	NCCLS M27-A	132	NA	0.001	0.002	NA	NA	NA	Clinical isolates
Jessup [101], USA, 2000	NCCLS-M38-P	30	0.001–0.002	NA	NA	NA	NA	NA	NA
Norris [102], Ohio, USA, 1999	NCCLS-M27-A	6	<0.06	NA	NA	NA	NA	NA	Clinical isolates
<i>CLSI/NCCLS method: isolates from clinically non-responding patients</i>									
Ebert [29], Germany/India, 2020	CLSI M38-A2	8	8	8*	8*	8*	8*	F397L	Skin
Sardana [18], India, 2018	CLSI M38-A2	11	NA	NA	0.125	NA	NA	NA	Skin
Pathania [47], India, 2018	CLSI M38-A2	29	0.015–16	0.063	8	NA	NA	NA	Skin, nails
Rudramurthy [11], India, 2018	CLSI M38-A2	5	2–16	8*	16*	6.06*	NA	F397L	Skin
Salehi [20], Iran, 2018	CLSI M38-A2	1 ^a	>32	NA	NA	NA	NA	L393F	Skin
Yamada [25], Switzerland, 2017	CLSI M38-A2	12 ^a	0.1–>12.8	3.2*	6.4*	NA	6.4*	F397L L393F L393S F393I F393V F415I F415S F415V H440Y	Clinical isolates
Ghannoum [49], USA, 2011	CLSI M38-A2	5 ^a	2–8	NA	NA	NA	NA	NA	Clinical isolates
Gupta [110], Canada, 2003	NCCLS-M27-A	18	0.003–2	0.003	0.003	0.003	0.003*	NA	Nails
EUCAST method: All isolates									
Markantonatou [77], Greece, 2020	EUCAST E.Def 9.3.1	8	<0.031–0.063	<0.031	0.063*	NA	<0.031*	NA	Clinical isolates

Table 2 (continued)

First author [Ref.], country, year	Method	Isolates, <i>n</i>	MIC range, mg/L	MIC ₅₀ , mg/L	MIC ₉₀ , mg/L	GM-MIC, mg/L	Modal MIC, mg/L	MIC for isolate with clinical resistance	Material
Arendrup [48], Denmark, 2020	EUCAST E.Def 11.0	WT 10 Mutant 8	Visual reading <0.008–0.5 Spec-90% <0.008–1 Spec-50% <0.008–0.5 Visual reading 0.06–>8 Spec-90% 0.03–>8 Spec-50% 0.016–>8	0.03 0.03 0.008 4 4 1	NA NA NA NA NA NA	NA NA NA NA NA NA	0.03 0.03 0.008 4 4 2	NA F397L L393S L393F I121M/V237I	Clinical isolates
Saunte [22], Denmark, 2019	EUCAST E.Def 11.0	WT 8 Mutant 12	Without CC 0.016–0.06 With CC 0.016–0.06 Without CC 0.125–>8 With CC 0.125–>8	0.03* 0.03* 8* 1*	0.06* 0.06* >8* >8*	0.028* 0.030* NA NA	0.03* 0.03* NA NA	NA F397L L393S L393F F415S I121M/V237I H440Y/F484Y	Clinical isolates
Dabas [70], India, 2017	EUCAST E.Def 9.2	5	0.06–8	0.5	8	0.991	0.5	NA	Skin
da Silva Barros [96], Brazil, 2006	EUCAST E.Dis 7.1	50	0.007–0.031	0.007*	0.015*	NA	0.007*	NA	Nails
<i>EUCAST method: Isolates from clinically non-responding patients</i>									
Arendrup [48], Denmark, 2020	EUCAST E.Def 11.0	8	Visual reading 0.06–>8 Spec-90% 0.03–>8 Spec-50% 0.016–>8	4 4 1	NA NA NA	NA NA NA	4 4 2	F397L L393S L393F I121M/V237I	Clinical isolates
Saunte [22], Denmark, 2019	EUCAST E.Def 11.0	12	Without CC 0.125–>8 With CC 0.125–>8	8* 1*	>8* >8*	NA NA	NA NA	F397L L393S L393F F415S I121M/V237I H440Y/F484Y	Clinical isolates
<i>Microdilution method unspecified: all isolates</i>									
Santos [103], Brazil, 2006	Microdilution	52	<0.031	<0.031	<0.031	NA	NA	NA	Nails
Bradley [113], USA, 1999	Microdilution	30	<0.001–0.015	NA	NA	NA	0.001*	NA	Nails
Korting [104], Germany, 1995	Microdilution	32	0.001–0.05	NA	NA	NA	0.01*	NA	Nails
<i>Microdilution method unspecified: isolates from clinically non-responding patients</i>									
Bradley [113], USA, 1999	Microdilution	30	<0.001–0.015	NA	NA	NA	0.001*	NA	Nails

MIC, minimal inhibitory concentration; MIC₅₀, MIC in which 50% of isolates were inhibited; MIC₉₀, MIC in which 90% of isolates were inhibited; G_{mean} , geometric mean; CLSI, Clinical Laboratory Standards Institute; NCCLS, National Committee for Clinical Laboratory Standards (former name of CLSI); NA, not available; WT, wild type; spec-90%/spec-50%, plate reading by spectrophotometer using 90%/50% inhibition endpoint; CC, medium supplemented with chloramphenicol & cycloheximide; NIDDM, non-insulin-dependent diabetes mellitus; mPDB medium, modified potato dextrose broth medium; RMP1 1640, Roswell Park Memorial Institute 1640. * Calculated in this study. ^a Suspected resistance because of high MIC, and/or SQLE, mutation.

tecting the following SQLE mutations: F397L, L393F, L393S, H440Y, F393I, F393V, F415I, F415S, F415V, I121M/V237I and H440Y/F484Y. Three studies using the CLSI reference standard proposed the following breakpoints for *T. rubrum* strains: TRB resistance was defined as $R \geq 1$ mg/L [18, 49], $R \geq 4$ mg/L [47], and one study proposed high TRB MIC at ≥ 2 mg/L [11]. One study using EUCAST protocol reported following WT-UL: WT-UL ≤ 0.125 mg/L (by visual and spec-90% endpoint reading) [48], WT-UL ≤ 0.03 mg/L (with spec-50% endpoint reading in plates containing chloramphenicol and cycloheximide [CC]) [48], $S \leq 0.06$ mg/L (with spec-50% endpoint reading in plates without CC) [48], and one study proposed ECOFF of 0.06 or 0.125 [22]. EUCAST has recently established a tentative ECOFF with the spec-50% endpoint reading of 0.03 mg/L for *T. rubrum* and E.Def 11.0 [43].

Discussion

The number of patients with clinical and laboratory-confirmed TRB-resistant dermatophytosis is rising [5, 22]. Until this autumn (2020), there were no official guidelines on how to interpret the results and furthermore, there is a huge intraspecies variation in reported MIC values for the most prevalent species *T. rubrum*, *T. mentagrophytes* and *T. interdigitale*. This variation can be due to different methods in AFST and technical issues related to testing, or it can reflect true differences in susceptibility due to acquisition of point mutations in the SQLE gene altering the binding site of TRB and conferring loss of susceptibility to TRB therapy [11, 12, 22, 28, 48].

However, clinical resistance does not always correspond to the *in vitro* resistance and vice versa. This is because several factors influence outcome of an infection including the severity of infection, the host immunity, the timing and dosing of the therapy, compliance and the susceptibility of the infecting organism. A newer study from 2018 investigated risk factors and performed AFST of both *T. rubrum* and *T. interdigitale* strains from 39 recurrent and clinically unresponsive tinea corporis and tinea cruris patients [47]. TRB MIC ranging from 0.015 to 16 mg/L were found, including 1 *T. interdigitale* and 3 *T. rubrum* isolates with MICs of >2 mg/L, which is only 10% of the total number of clinically resistant cases. The authors concluded that most of the strains were not drug resistant, and the increase in MIC was not the only factor responsible for clinical resistance in dermatophytosis.

The management of TRB-resistant infections is challenging as few alternatives are available. There is compelling evidence that itraconazole is the preferred agent for TRB-resistant *Trichophyton* infections [5, 29, 33, 50, 51]. The optimal dosing regimen and treatment duration, however, have not been established. Consequently, practices vary, and elevated doses above 200–400 mg daily are sometimes adopted for prolonged duration despite lack of scientific rationale or clinical validation [52, 53]. The current prevailing sentiment in India is to reduce unnecessary prescription of itraconazole as an attempt to avoid emergence of itraconazole resistance in *Trichophyton* as emergence of multidrug resistant infections would have grave consequences. Azole resistant dermatophytes have previously been reported [54], and few cases have been found to be resistant to griseofulvin since the 1960s [55, 56]. Cross-resistance within the same antifungal group has also been noted for example, resistance to other allylamines (e.g., butenafine, naftifine) in TRB-resistant isolates [57]. The antifungal mode of action differs, as azoles and TRB both inhibit the ergosterol biosynthesis (but inhibit different enzymes), and griseofulvin interferes with the microtubule formation [58, 59]. Nevertheless, multidrug resistance to several drug classes has been reported after serial passages on drug containing agars *in vitro* [60]. Moreover, multidrug resistance has been sporadically reported in the clinical setting of TRB-resistant *T. mentagrophytes/interdigitale* infection in India [59]. The Indian epidemic is probably somewhat unique and explained by the emergence of a new and more virulent *T. mentagrophytes* subspecies (genotype VIII) in response to a high use of antifungal agents potentially facilitated/enhanced by an animal reservoir and challenges in infection control due to crowded housing conditions. However, TRB resistance is increasing elsewhere despite the lack of routine testing and the prevalence of multidrug resistance largely unknown [10, 18, 29, 37].

Interpretative clinical breakpoint values were not established until recently. CLSI has noted that most TRB MICs are ≤ 0.25 mg/L for dermatophytes [61]. This value is, however, several fold higher than the recommended CLSI MIC range for the *T. mentagrophytes* quality control strain (0.002–0.008 mg/L) raising a concern that not all isolates with a MIC up towards 0.25 mg/L may be without resistance mechanisms and clinically susceptible. Of note, some studies using the CLSI method suggest even higher breakpoints for resistance for *T. interdigitale* between ≥ 1 and ≥ 4 mg/L [11, 12, 46, 47], *T. mentagrophytes* between ≥ 1 and ≥ 4 mg/L [18, 47], *T. rubrum* between ≥ 1 and ≥ 4 mg/L [18, 47, 49] and *T. mentagrophytes-T. interdigitale*

complex WT-UL ≤ 8 mg/L [5]. Thus, further standardisation and establishment of CLSI breakpoints are urgently needed.

Recently, EUCAST developed a reference method that allows an automated endpoint reading and provides quality control ranges for several control strains [43]. This method will facilitate improved inter-laboratory reproducibility. Moreover, tentative ECOFFs were established which allow classification of isolates as wild type or non-wild type. Several aspects are considered and integrated into the process of setting breakpoints for antifungal agents. These include dosing information, MIC distributions, ECOFF setting, pharmacokinetics, pharmacodynamics and clinical outcome data. Until such data is available, adoption of the EUCAST method with the tentative ECOFFs will allow guidance for TRB, itraconazole and voriconazole use against *T. rubrum* and *T. interdigitale*. Wildtype isolates are “normal” and will respond “normally” to treatment, meaning the clinician can expect the typical outcome for treatment with a specific drug-bug combination. Non-wild-type isolates may or may not respond to therapy depending on the degree of MIC elevation and the drug exposure that can be obtained. A rule of thumb is nevertheless, that the clinician should consider if alternative therapy would be warranted. Most important is to realise that MICs, ECOFFs and breakpoints are method dependent, and just like a distance is a different number if measured in metres or feet, MIC values across methods will vary and can only be interpreted correctly by the method-specific associated ECOFFs and breakpoints. This is also true for susceptibility results obtained by commercial test systems. Such results are only reliable if the method is correctly standardised against the reference method from which the breakpoints are adopted.

Conclusion

Overall, high TRB MICs were reported in the dermatology literature for *T. mentagrophytes*, *T. interdigitale* and *T. rubrum* and more than 15 individual target protein alterations have been identified in isolates from treatment-resistant cases [62–65]. Clinically, the number of patients with resistant dermatophytosis is also rising, supporting the increasing clinical relevance of AFST in non-responding cases and where resistance is prevalent [5, 10, 22]. AFST is unfortunately not considered a routine test but recent progress allows for standardisation and a broader implementation of dermatophyte suscep-

tibility testing. Hopefully, this will help bring AFST a step closer to becoming a routine test and guide, as susceptibility testing is for so many other infections.

Key Message

Clinical suspected terbinafine resistance can be mycologically proven by microdilution methods. Recent progress allows for standardisation.

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Statement of Ethics

Written informed consent was obtained from the patients for publication of the details of their medical case and any accompanying images.

Conflict of Interest Statement

M.C. Arendrup has, outside the current work, received research grants/contract work (paid to the SSI) from Amplyx, Basilea, Cidara, F2G, Gilead, Novabiotics, Scynexis and T2Biosystems and speaker honoraria (personal fee) from Astellas, Gilead, MSD, and SEGES over the past 5 years. She is the current chairman of the EUCAST-AFST.

S. Verma has delivered one lecture and has attended an “Ad Board meeting” once, both organised by Janssen Pharmaceuticals, India, for which he received a honorarium.

D.M.L. Saunte was a paid consultant for advisory board meetings by AbbVie, Janssen, Sanofi, Leo Pharma and received speaker's honoraria and/or received grants from the following companies: Abbvie, Desitin, Pfizer, Galderma, Novartis and Leo Pharma during the last 5 years.

J.J. Shen has no conflict of interests.

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Author Contributions

D.M.L. Saunte initiated the article, J.J. Shen and D.M.L. Saunte performed the systematic review of included articles. All authors wrote and critically reviewed the manuscript. All authors have accepted the final version of the manuscript.

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